By lemma $d(x,w)^2 = d(x,z)^2 + d(z,w)^2 \pm \Psi$

and $d(z,w)^2 = |h_+(w) - \underset{h_+(x)}{\overset{\shortparallel}{h_+(z)}}|^2 \pm \Psi$

(in $\mathbb{R}\times X$, $d_{\mathbb{R}\times X}(F(x),F(w))^2 = |h_+(w) - h_+(x)|^2 + d_X(x',w')^2$

using lemma twice more we get

$$d(x',w')^2 = d(x,z)^2 \pm \Psi$$

$$\Rightarrow \quad d(F(x),F(w)) = d(x,w) \pm \Psi$$

Can also see that $F(B_{R/8}(p))$ is Ψ-dense in $B_{R/8}((0,p')) \subset \mathbb{R}\times X$. $\square$

<u>Thm</u> (Almost splitting theorem) Assume $(*)$ then $\exists$ (length space) X, $x \in X$ s.t.

$$d_{GH}(B_R(p), B_R((0,x))) < \Psi(\delta, L^{-1}, \varepsilon \mid n, R)$$

$(0,x) \in \mathbb{R}\times X$.

<u>More general setup</u>

<u>Defn</u> A (k,ε)-splitting map is a smooth $u : B_r(p) \longrightarrow \mathbb{R}^k$ $(B_r(p) \subset M^n)$ s.t.

1) $\Delta u = 0$
2) $|\nabla u|_* < 1+\varepsilon$ $\quad$ ($|\nabla u|_*$ = operator norm of $\nabla u : TM \to T\mathbb{R}^n$)
3) $\fint_{B_r(p)} |\langle \nabla u_i, \nabla u_j\rangle - \delta_{ij}|^2 < \varepsilon^2$
4) $\fint_{B_r(p)} r^2 |\nabla^2 u_i|^2 < \varepsilon^2$

<u>Note</u>: definition is scale invariant:

if $\tilde{g} = \lambda^2 g$, $\tilde{u} = \lambda u$ on $B_{\lambda r}(\tilde{p}) \subset (M^n, \lambda^2 g)$

$$\langle \tilde{\nabla} \tilde{u}_i, \tilde{\nabla} \tilde{u}_j \rangle_{\tilde{g}} = \langle \nabla u_i, \nabla u_j \rangle_g$$

$$\lambda r |\tilde{\nabla}^2 \tilde{u}_i|_{\tilde{g}} = \lambda r \lambda^{-1} |\nabla^2 u_i|_g$$

<u>Lemma</u> Given $\varepsilon, R > 0$, $\exists\, \delta = \delta(n, \varepsilon, R) > 0$ s.t.

if $\mathrm{Ric} \geq -(n-1)\delta$ then

(1) If $u : B_{2R}(p) \to \mathbb{R}^k$ is a (k, δ)-splitting map

then $\exists\, f : B_R(p) \to u^{-1}(u(p))$ s.t.

$$(u, f) : B_R(p) \to \mathbb{R}^k \times u^{-1}(u(p))$$

is an ε-GH approximation

(2) If $d_{GH}\left(B_{\delta^{-1}}(p), B_{\delta^{-1}}((0,y)) \right) < \delta$

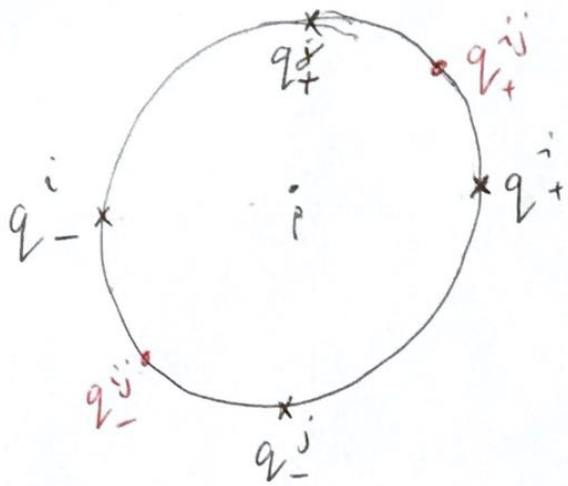
for $(0,y) \in \mathbb{R}^k \times Y$ then

$\exists\, (k, \varepsilon)$-splitting map $u : B_R(p) \to \mathbb{R}^k$

<u>Proof</u> is generalization of ideas in almost splitting thm.

(1) Define f to be nearest point projection to $u^{-1}(u(p))$. Use segment ineq. to convert integral bounds for $\nabla u, \nabla^2 u$ to distance estimates.

(2) Find points $q_\pm^1, \dots, q_\pm^k$ approximating $(\pm \delta^{-1} e_i, y)$, $e_i \in \mathbb{R}^k$ standard unit vectors



Corresponding $h_\pm^1, \dots, h_\pm^k$ satisfy $\fint_{B_R(p)} \big| |\nabla h_\pm^i|^2 - 1 \big| < \Psi$

$$\fint_{B_R(p)} |\nabla^2 h_\pm^i|^2 < \Psi$$

To see $\fint_{B_R(p)} |\langle \nabla h_+^i, \nabla h_+^j \rangle| < \Psi$ if $i \neq j$

use $q_\pm^{ij} = \frac{1}{\sqrt{2}}(q_\pm^i + q_\pm^j)$ & corresponding $h_\pm^{ij}$

show that $h_\pm^{ij} = \dfrac{h_\pm^i + h_\pm^j}{\sqrt{2}} \pm \Psi$

and use $\fint \big| |\nabla h_\pm^{ij}|^2 - 1 \big|^2 < \Psi$ $\square$

<u>Theorem</u> (Volume convergence, Colding)

Let $\mathrm{Ric}_{M^n} \geq -(n-1)\delta$

$$d_{GH}(B_R(p), B_R(0)) < \delta \quad , \quad \begin{matrix} p \in M^n \\ 0 \in \mathbb{R}^n \end{matrix}$$

$$\Rightarrow \mathrm{Vol}(B_R(p)) \geq (1 - \Psi(\delta|n))\, \mathrm{Vol}(B_R(0))$$

<u>Proof</u> <u>Step 1</u> Reduce to case where

$$d_{GH}(B_{\delta^{-1}}(p), B_{\delta^{-1}}(0)) < \delta$$

by filling most of $B_R(p)$ by disjoint balls $B_{r_i}(x_i)$ s.t. $r_i \ll d(x_i, \partial B_R(0))$

<u>Step 2</u> We have (n,ε)-splitting $u: B_R(p) \to \mathbb{R}^n$

$\varepsilon = \Psi(\delta|n,R)$

<u>Claim</u> $\exists\, V \subset B_R(p)$ s.t.

$$\mathrm{Vol}(V) \geq (1 - \Psi)\, \mathrm{Vol}(B_R(p))$$

s.t. if $z \in V$, $B_r(z) \subset B_R(p)$

then u is (n, ε')-splitting on $B_r(z)$

$$\varepsilon' = \Psi(\delta|n,R)$$

<u>Proof of Claim</u>: Let $\mathcal{B} = \left\{ B_r(x) \subset B_R(p) : \begin{array}{l} u \text{ is not } (n, \varepsilon')\text{-splitting} \\ \text{on } B_r(x) \end{array} \right\}$

i.e. $\displaystyle\fint_{B_r(x)} r^2 |\nabla^2 u_i|^2 + |\langle \nabla u_i, \nabla u_j \rangle - \delta_{ij}|^2 \geq \varepsilon'^2$ for $B_r(x) \in \mathcal{B}$

Vitali covering lemma

$\Rightarrow$ can find collection $\{B_{r_i}(x_i)\} \subset \mathcal{B}$ of disjoint balls s.t.

$$\bigcup_{B \in \mathcal{B}} B \subset \bigcup_i B_{5r_i}(x_i)$$

Then

$$\sum_i \varepsilon'^2 \frac{\mathrm{Vol}(B_{r_i}(x_i))}{\mathrm{Vol}(B_R(p))} \leq \sum_i \frac{1}{\mathrm{Vol}(B_R(p))} \int_{B_{r_i}(x_i)} r^2 |\nabla^2 u_i|^2 + |\ldots|^2 < 2\varepsilon$$

$$\mathrm{Vol}\Big(\bigcup_{B \in \mathcal{B}} B\Big) \leq \sum_i \underbrace{\frac{\mathrm{Vol}\, B_{5r_i}(x_i)}{\mathrm{Vol}\, B_{r_i}(x_i)}}_{\leq C} \mathrm{Vol}\, B_{r_i}(x_i)$$

$$\leq C \cdot \varepsilon'^{-2} \mathrm{Vol}(B_R(p)) \cdot 2\varepsilon$$

can let $\varepsilon' = \varepsilon^{1/2} = \Psi$

$$\Rightarrow \mathrm{Vol}\Big(\underbrace{B_R(p) \setminus \bigcup_{B \in \mathcal{B}} B}_{V}\Big) \geq (1 - \Psi)\, \mathrm{Vol}(B_R(p))$$